IIT ROORKEE

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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-52

DC Charging (CCS2)-III

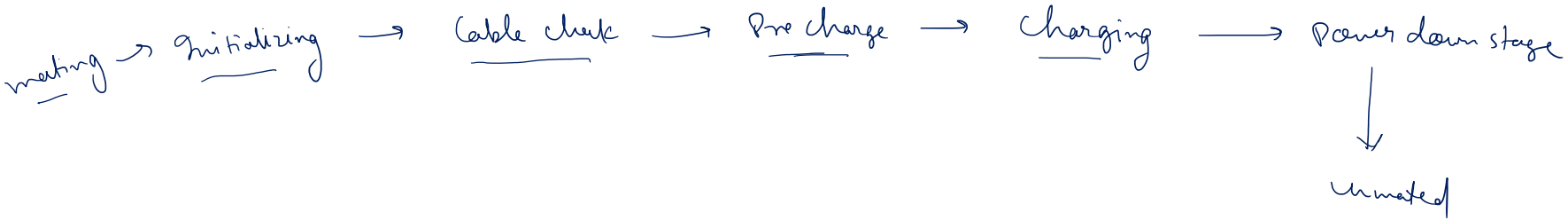
Dr. Apurv Kumar Yadav

Department of Electrical Engineering

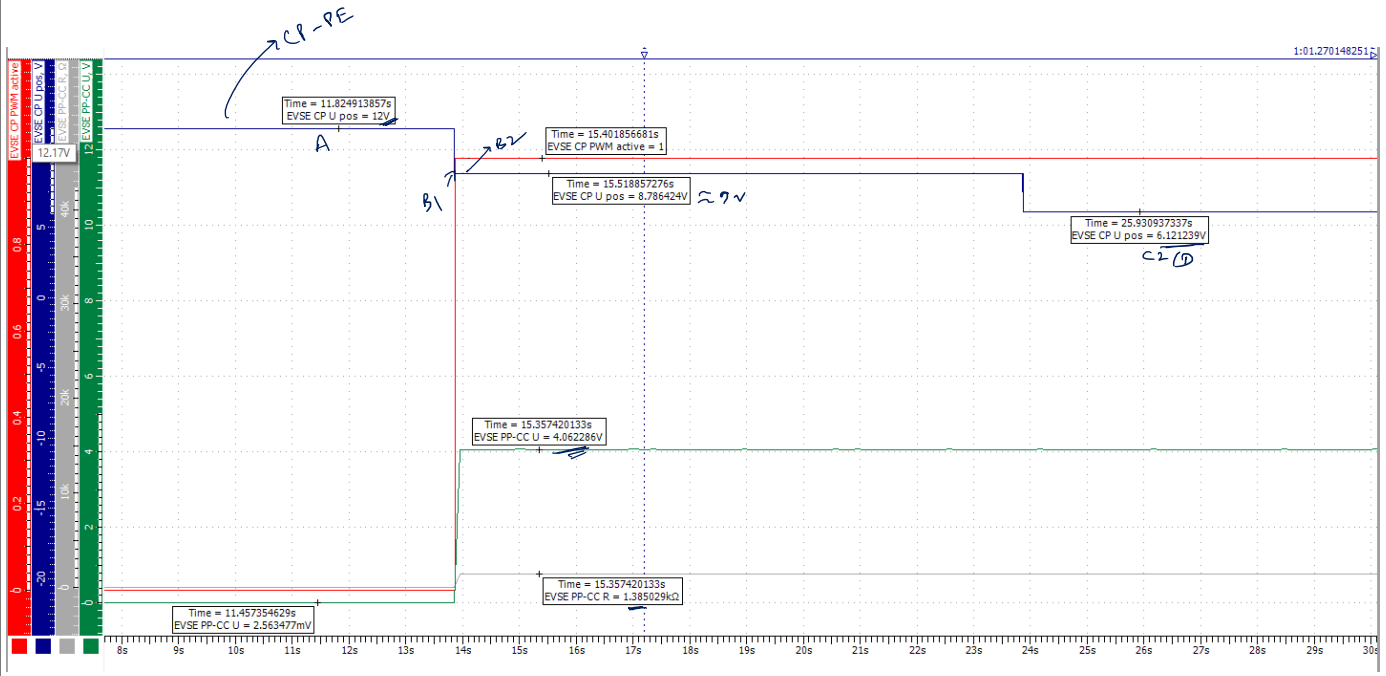


Recap: DC Charging Sequence with CCS2 Charging

- CCS2 charging sequence is being studied.
- Charging sequence is given in IEC 61851-23 and also being included in AIS-138



DC CCS2 Charging Sequence



A → B1 → B2 → C/D → B2 → B1 → A

PP-PE → 1500 Ω for DC CCS2

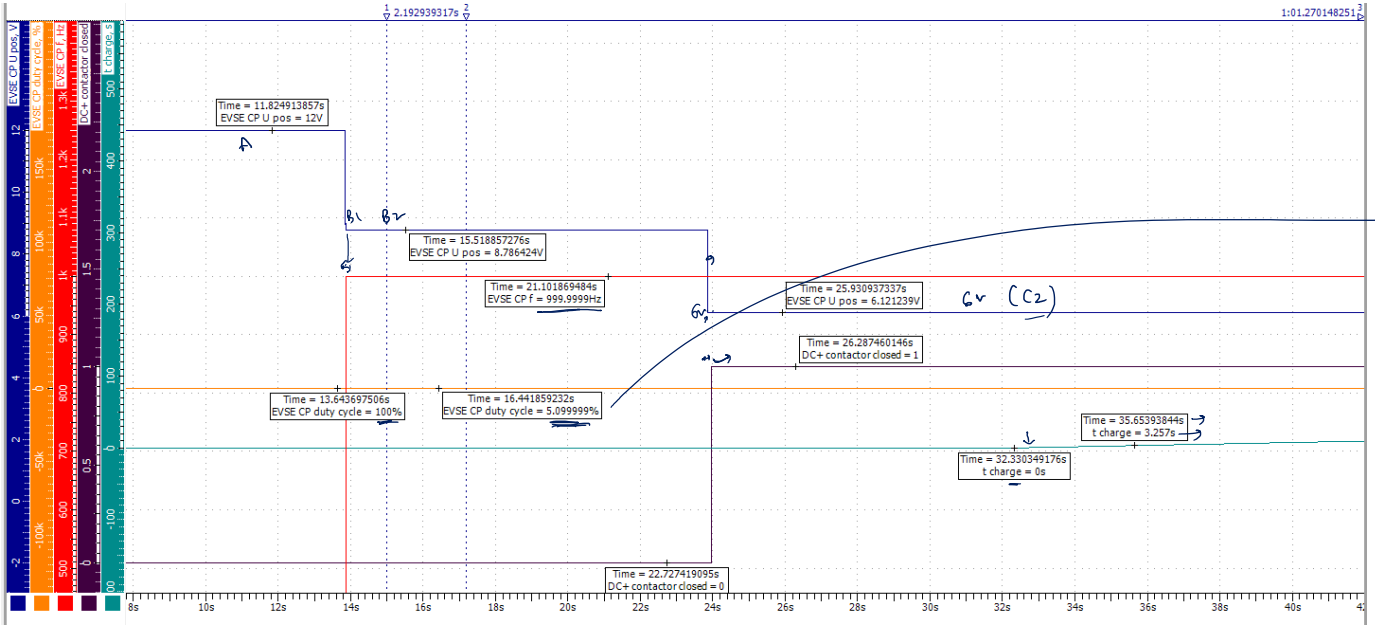
$(1100 - 2460\,\Omega) \parallel (4700)$

$= 891.379\,\Omega - 1588.732\,\Omega$

PP-PE $\left(\frac{1385}{1385+330}\right) \times 5V = 4.03V$

Courtesy: Data taken from Charging Discovery System of Keysight installed in the Department of Electrical Engineering, IIT Roorkee

DC CCS2 Charging Sequence

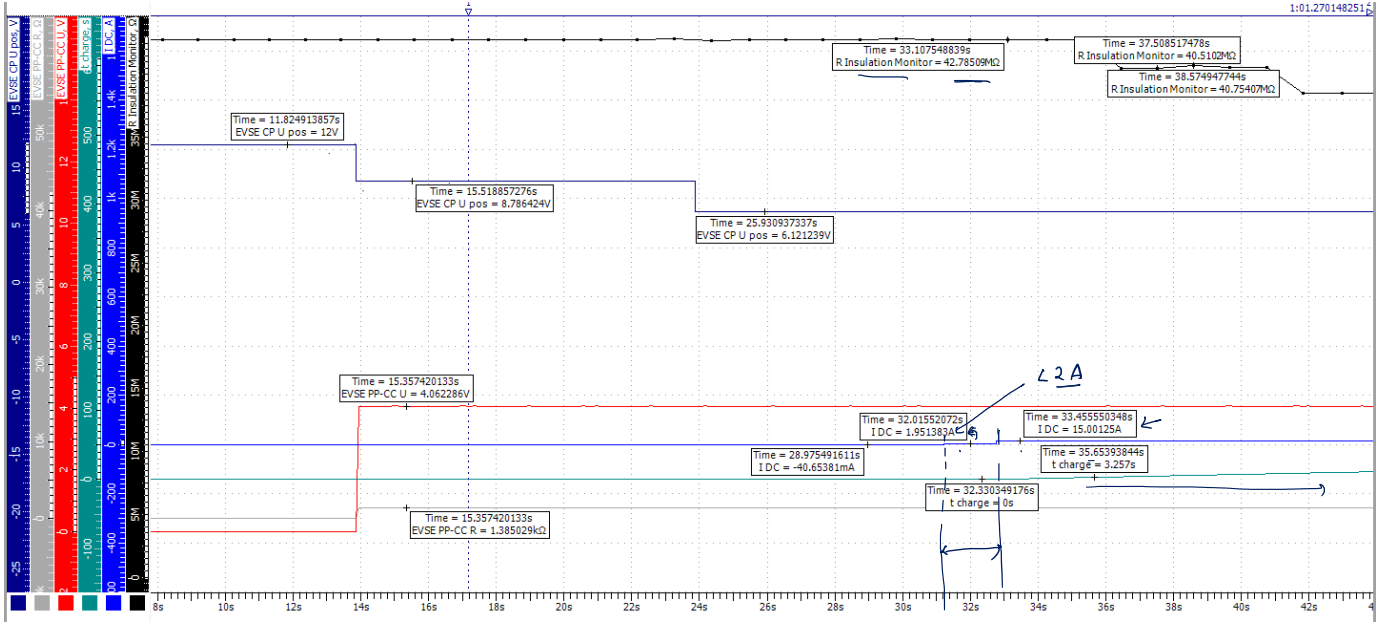


in the range 3-7%

→ HLC using PLC based Communication

Courtesy: Data taken from Charging Discovery System of Keysight installed in the Department of Electrical Engineering, IIT Roorkee

DC CCS2 Charging Sequence

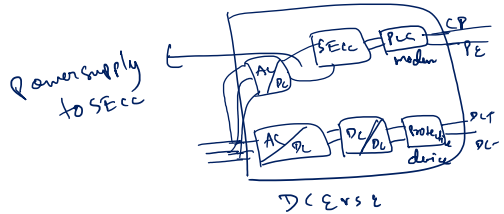


DC - DC) Rinsulation > 100kΩ

Courtesy: Data taken from Charging Discovery System of Keysight installed in the Department of Electrical Engineering, IIT Roorkee

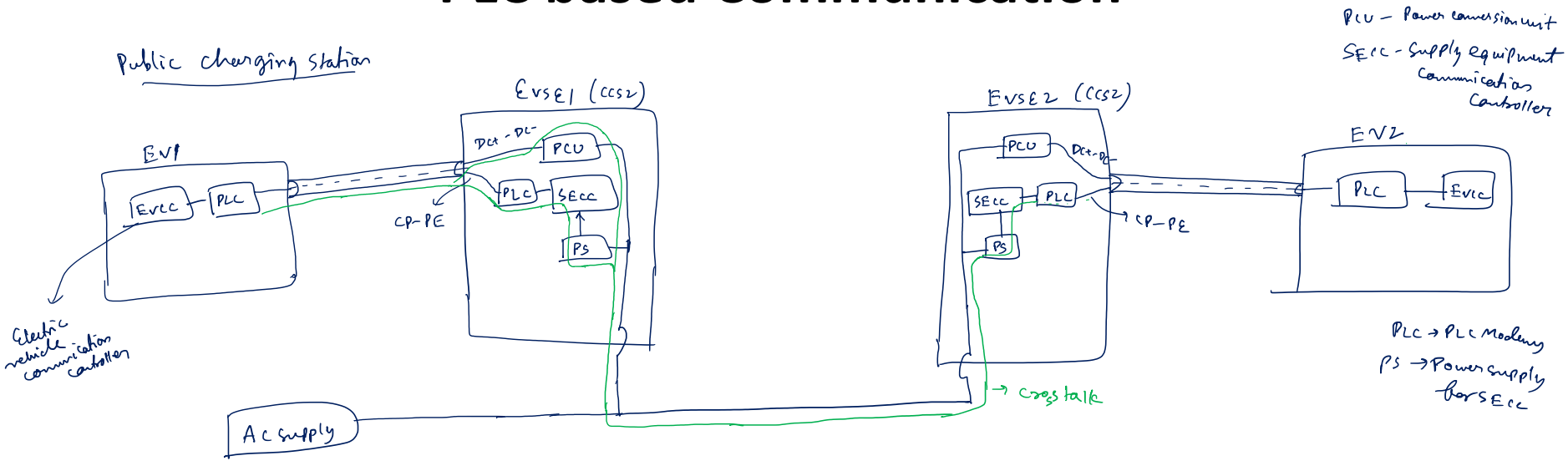
Recap: PLC based Communication

- The IEC 61851-23 standard, details the requirements for DC charging stations, specifically mentions the use of PLC for the High-Level Communication.
- HomePlug Green PHY (HPGP) based PLC is being used for DC charging of CCS2 with data being transmitted over 2–28 MHz wideband.
- The HPGP digital data transmission rate is 4.5 to 10 Mb/s.
- **Correct PLC modems of EV and EVSE need to be logically connected** to perform high level communication.



1. Ramasamy, D. Possible Hardware Architectures for Power Line Communication in Automotive V2G Applications. *J. Inst. Eng. India Ser. B* **104**, 813–819 (2023). https://doi.org/10.1007/s40031-023-00880-5.
2. Bahrami, Ali. (2020). EV Charging Definitions, Modes, Levels, Communication Protocols and Applied Standards. 10.13140/RG.2.2.15844.53123/11.

PLC based Communication





Possibilities are

✓ → PLC modem of EV1 can get logically connected to PLC modem of EVSE1

✗ → PLC modem of EV1 can get logically connected with PLC modem of EVSE2

✗ → PLC modem of EV2 can get logically connected with PLC modem of EVSE1

✓ → PLC modem of EV2 can get logically connected with PLC modem of EVSE2

SLAC Procedure

- The Signal Level Attenuation Characterization (SLAC) is a protocol to ensure logical connection of PLC modems of EV and EVSE (that are physically connected to each other).
- SLAC is defined in HomePlug Green PHY v1.1.1 specification.
- SLAC is a protocol which measures the attenuation between two Power Line Communication (PLC) modules.
- SLAC procedure is performed after the charging enters the B2 state.
- **PLC modems of EV and EVSE are logically connected** (to perform high level communication) after the SLAC procedure.
- It is a part of datalink layer in High level communication.

Source:
[1] Shin M, Kim H, Kim H, Jang H. Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards. *Applied Sciences*. 2016; 6(6):165. https://doi.org/10.3390/app6060165
[2] detailed SLAC procedure is given in ISO 15118-3 and DIN SPEC 70121 standards

SLAC Procedure Cont.

- If there are several EV's that are connected to charging stations nearby, there can occur crosstalk in between.
- To ensure communication only between physically connected EV and EVSE, SLAC procedure is performed.
- PLC modules which show the lowest attenuation to each other are physically connected.

Source:
[1] Shin M, Kim H, Kim H, Jang H. Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards. *Applied Sciences*. 2016; 6(6):165. https://doi.org/10.3390/app6060165
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SLAC Procedure Cont.

- Protocol to setup the communication channel between EV and EVSE.
- It is given in detailed in ISO 15118-3 and DIN SPEC 70121 standards.
- EV initiates by sending broadcast message to all the EVSEs present in the charging station.
- It includes following steps
  - Parameter exchange/setup
  - Discovery of PLC nodes
  - Matching decision
  - Validation (optional)
  - Key exchange
  - Logical connection

Source:
[1] Shin M, Kim H, Kim H, Jang H. Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards. *Applied Sciences*. 2016; 6(6):165. https://doi.org/10.3390/app6060165
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SLAC Procedure Cont.

- Parameter exchange/setup
  - **CM_SLAC_PARM.REQ** is sent by EV in the form of broadcast message. (Request)
  - All EVSEs are listening to this request will send a confirm message **CM_SLAC_PARM.CNF** .
  - The confirm message contains the number of “sounding or attenuation indication” signals which should be sent by the EV.
  - It sets up the SLAC parameters such as RunID, allowed frequency, timings, attenuation thresholds, etc.

Source:
[1] Shin M, Kim H, Kim H, Jang H. Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards. *Applied Sciences*. 2016; 6(6):165. https://doi.org/10.3390/app6060165
[2] detailed SLAC procedure is given in ISO 15118-3 and DIN SPEC 70121 standards

SLAC Procedure Cont.

- Discovery of PLC nodes
  - EV broadcasts 3 times the **CM_START_ATTEN_CHAR.IND** message. It act as a wakeup call for all the EVSEs which sends the confirm messages.
  - The EVSE will get ready to receive the sounding or attenuation indication messages.
  - The EV broadcasts/sends the preferred amount of sounding messages (normally 10) with each “sounding signal” comprises of 58 different frequency groups.
  - Each frequency group corresponds to the narrow band of frequencies with in the PLC frequency band.

Source:
[1] Shin M, Kim H, Kim H, Jang H. Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards. *Applied Sciences*. 2016; 6(6):165. https://doi.org/10.3390/app6060165
[2] detailed SLAC procedure is given in ISO 15118-3 and DIN SPEC 70121 standards

SLAC Procedure Cont.

- Discovery of PLC nodes
  - For each **CM_MNBC_SOUND.IND** message, the PLC modem on EVSE side will produce a **CM_PROFILE.IND** message.
  - After receiving all the 10 messages, the PLC EVSE sends the **CM_ATTN_CHAR.IND** message containing the average attenuation values for 58 frequency groups.
  - The EV acknowledges the attenuation profile result and starts the analysis for matching decision.

Source:
[1] Shin M, Kim H, Kim H, Jang H. Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards. *Applied Sciences*. 2016; 6(6):165. https://doi.org/10.3390/app6060165
[2] detailed SLAC procedure is given in ISO 15118-3 and DIN SPEC 70121 standards

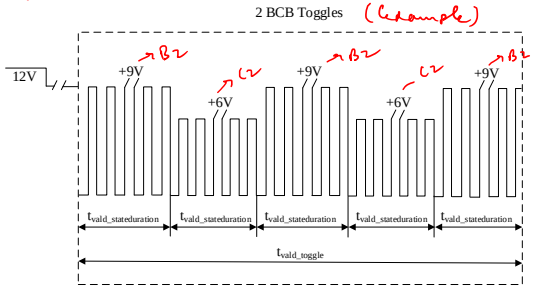
SLAC Procedure Cont.

- Matching Decision
  - If more than one EVSEs has answered the requests of the EV (due to cross-talk), then the PLC Modem of EV decides to which PLC Modem of EVSE to be connected to.
  - PLC Modem of EVSE which has the least attenuation value will be chosen for logical connection, if the results are well defined (for e.g, if only 1 EVSE has responded).
  - If the result is undecided, then a validation process is required, otherwise the EV can advance with the matching process by sending **CM_ATTEN_CHAR_RES** to the chosen EVSE.

Source:
[1] Shin M, Kim H, Kim H, Jang H. Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards. *Applied Sciences*. 2016; 6(6):165. https://doi.org/10.3390/app6060165
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SLAC Procedure Cont.

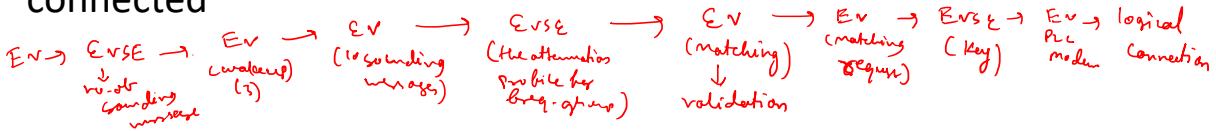
- Validation (optional)
  - EV sends a **CM_VALID_CPLT.REQ** to start the validation process which is confirmed by EVSE by sending **CM_VALID_CPLT.CNF** message.
  - EV toggles the charging states by closing and re-opening S2 switch (BCB toggle).
  - EV again sends a **CM_VALID_CPLT.REQ**
  - The EVSE confirms the message with the number of detected BCB toggles, which decides which EVSE to be connected to (it will be that EVSE which is physically connected by CCS2 charger plug).



Source:
[1] Shin M, Kim H, Kim H, Jang H. Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards. *Applied Sciences*. 2016; 6(6):165. https://doi.org/10.3390/app6060165
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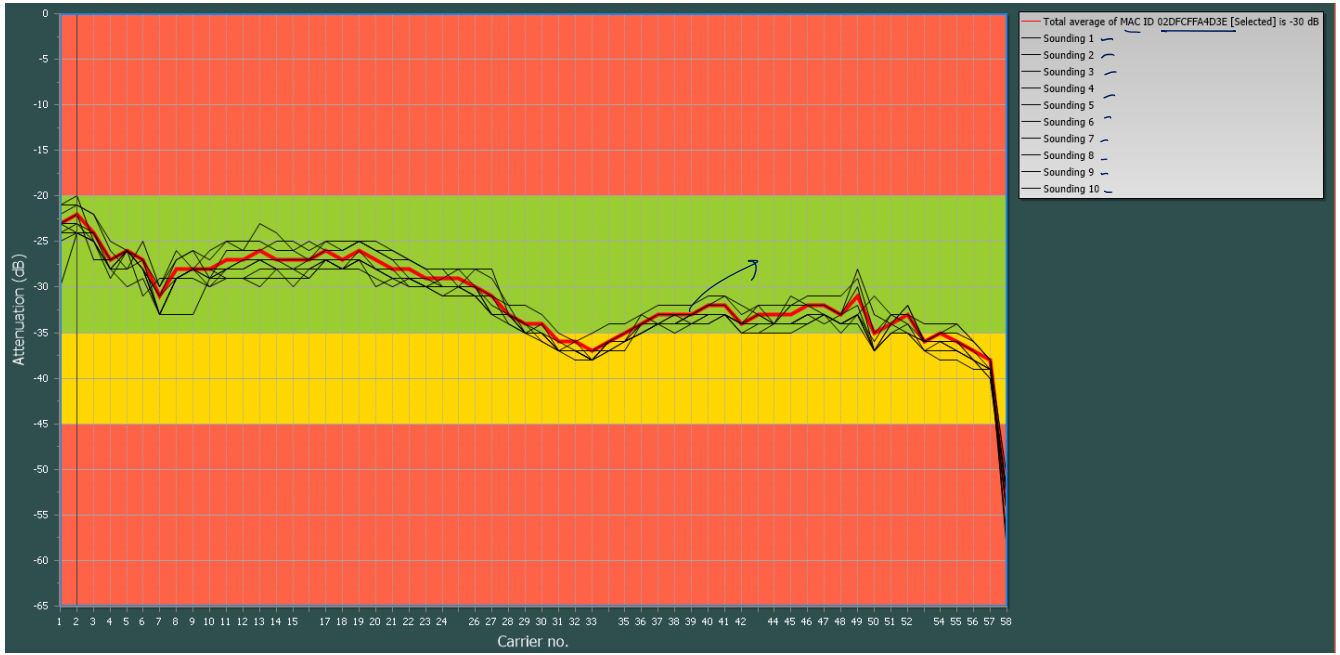
SLAC Procedure Cont.

- Key exchange
  - After the EVSE is chosen, the EV informs EVSE by sending a **CM_SLAC_MATCH.REQ** message to the EVSE.
  - The EVSE confirms with **CM_SLAC_MATCH.CNF** message containing the logical network membership key (NMK) and the network ID (NID).
  - The EV sets the received keys in its PLC modem using the **CM_SET_KEY.REQ** which is confirmed by the PLC modem.
  - PLC Modem gets automatically rebooted.
  - After the reboot, the PLC modems of EV and EVSE are logically connected



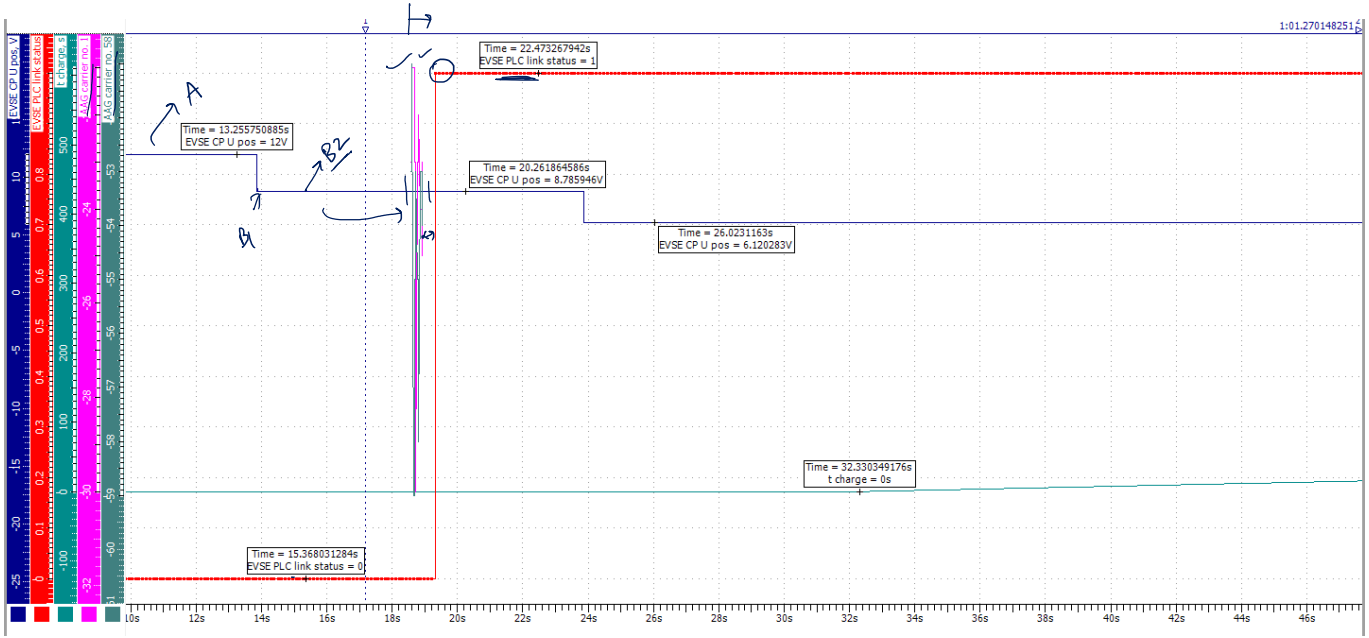
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[1] Shin M, Kim H, Kim H, Jang H. Building an Interoperability Test System for Electric Vehicle Chargers Based on ISO/IEC 15118 and IEC 61850 Standards. *Applied Sciences*. 2016; 6(6):165. https://doi.org/10.3390/app6060165
[2] detailed SLAC procedure is given in ISO 15118-3 and DIN SPEC 70121 standards

SLAC Procedure Cont.



Courtesy: Data taken from Charging Discovery System of Keysight installed in the Department of Electrical Engineering, IIT Roorkee

SLAC Procedure Cont.



PLC modems are logically connected

Courtesy: Data taken from Charging Discovery System of Keysight installed in the Department of Electrical Engineering, IIT Roorkee

SLAAC

- It is Stateless Address Autoconfiguration (SLAAC).
- IPv6 is autoconfigured with the assignment of local-link address or IP address to the EV and EVSE PLC Modems.

Source:
https://www.charin.global/media/pages/technology/ccs-specification/42a9d61e04-1626949173/design_guide_combined_charging_system_v7.pdf

Thank You